БЧХ - коды и РС - коды

Боуз, Рой-Чоудхури, Хаквингем

Рид, Соломон → границы символов $d \le r+1 = n-k+1$

1 —
2 Код с проверкой на чётность
3 код Хэмминга
4 расширенный код Хэмминга (7,4)

$$H = \begin{bmatrix} 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{bmatrix}$$

$2^3 = 8$

$1+x+x^3 = p(x)$

Пенсон 9

$GF(2) \to GF(2^3)$

1	0	0
x	α^0	1
$1+x$	α	x
x^2	α^2	x^2
x^2+1	α^3	$1+x$
x^2+x	α^4	$x+x^2$
x^2+x+1	α^5	$1+x+x^2$
	α^6	$1+x^2$

$$\begin{array}{r|l} x^3 & x^3+x+1 \\ \hline -\ x^3+x+1 & 1 \\ \hline x+1 & \end{array}$$

$$H = (1\ \alpha\ \alpha^2\ \alpha^3\ \alpha^4\ \alpha^5\ \alpha^6)$$

$$\begin{pmatrix} 1 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 \end{pmatrix}$$

$(15,11)$ $\quad r = n-k$

$H = (1\ \alpha\ \alpha^2 \dots \alpha^{14})$

$GF(2) \to GF(2^4)$

$p(x) = 1+x+x^4$

$1+x+x^3$

$c(x) + e(x) = b(x)$

$$S = \sum_{i=0}^{n-1} b_i \alpha^i = \sum_{i=0}^{n-1} (c_i + e_i)\alpha^i = \sum_{i=0}^{n-1} e_i \alpha^i = e(\alpha)$$

В позиции i — ошибка $\quad e(x) = x^i$, $S = \alpha^i$

$e(x) = x^i + x^j = i \ne j$

$$\begin{cases} S_1 = \alpha^i + \alpha^j \\ S_2 = f(\alpha^i) + f(\alpha^j) \end{cases}$$

$H = (1\ \alpha\ \alpha^2 \dots \alpha^{14})$

$(\beta_1, \beta_2 \dots \beta_{15})$

$\beta_i \in GF(2^m)$

$f(\alpha) = c\cdot\alpha$

$f(\alpha) = \alpha^2$

$\vdots$

$f(\alpha) = \alpha^3$

$GF(2)$

$b(x^2) = b^2(x)$

$1^2 = 1, \quad 0^2 = 0$

$$H = \begin{pmatrix} 1 & \alpha & \alpha^2 & \alpha^3 & \alpha^4 & \alpha^5 & \alpha^6 & \alpha^7 & \alpha^8 & \alpha^9 & \alpha^{10} & \alpha^{11} & \alpha^{12} \\ 1 & \alpha^3 & \alpha^6 & \alpha^9 & \alpha^{12} & 1 & \alpha^3 & \alpha^6 & \dots & & & & \alpha^6 \alpha^9 \alpha^{12} \end{pmatrix}$$

i, j – позиции ошибки $\quad x = \alpha^i, \; y = \alpha^j$

$$\begin{cases} x + y = S_1 \\ x^3 + y^3 = S_2 \end{cases}$$

при одной или двух ошибк

$S_1 \neq 0$

$$\frac{x^3+y^3}{x+y} = \frac{S_2}{S_1} \qquad x^3+y^3 = (x+y)(x^2+y^2+xy)$$

$$x^2+y^2+xy = \frac{S_2}{S_1}$$

$$(a+b)^2 = a^2+b^2+2ab = a^2+b^2$$

$$x+y = S_1 \qquad (x+y)^2 = S_1^2$$

$$x^2+y^2 = S_1^2$$

$$S_1^2 + xy = \frac{S_2}{S_1} \qquad \begin{cases} xy = \frac{S_2}{S_1} - S_1^2 \\ x+y = S_1 \end{cases}$$

по т. Виета

$$\lambda_1 = S_1$$

$$\lambda_2 = \frac{S_2}{S_1} - S_1^2$$

$GF(2) \longrightarrow GF(2^4) \qquad p(x) = 1 + x + x^4$

α	полином	двоичное
0	1	0001
1	x	0010
2	x^2	0100
3	x^3	1000
4	$1+x$	0011
5	$x+x^2$	0110
6	x^2+x^3	1100
7	$1+x+x^3$	1011
8	$1+x^2$	0101
9	$x+x^3$	1010
10	$1+x+x^2$	0111
11	$x+x^2+x^3$	1110
12	$1+x+x^2+x^3$	1111
13	$1+x^2+x^3$	1101
14	$1+x^3$	1001

$$H \quad \begin{array}{c} 000100 \\ 001001 \\ 010011 \\ 100010 \\ \hline 011110 \\ 001010 \\ 000110 \\ 100011 \end{array} \quad \begin{array}{c} \longrightarrow \\ \\ \\ \\ \\ \\ \longrightarrow \\ \\ \end{array}$$

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
1	1	1	0	0	1	1	0	0	0	1	1	0	1	1
1	1	0	0	0	1	1	0	0	1	1	0	1	1	1

$S_1 = \alpha^2 + \alpha^9 = \alpha^{11}$ $\qquad$ $S_2 = \alpha^6 + \alpha^{12} = \alpha^4$

$$\begin{cases} xy = \dfrac{S_2}{S_1} - S_1^2 = \sigma^2 \\ x + y = S_1 = \sigma^1 \end{cases}$$

$$\frac{\alpha^4}{\alpha^{11}} + (\alpha^{11})^2 = \frac{1}{\alpha^7} + \alpha^7 = ; \quad \alpha^8 + \alpha^7 = \alpha^{11}$$

$$= \frac{1+\alpha^{14}}{\alpha^7} = \frac{\alpha^3}{\alpha^7} = \alpha^{-4}$$

$$\sigma^2 = \alpha^{11}\sigma^1 = \alpha^{11}$$

$$z^2 + \alpha^{11} z + \alpha^{11} = 0 \qquad \alpha^2 \quad \alpha^9$$

$$(\alpha^2)^2 + \alpha^{11}\alpha^2 + \alpha^{11} = \alpha^4 + \alpha^{13} + \alpha^{11} = 1 + x + 1 + x^2 + x^3 + x + x^2 + x^3 = 0$$

$$(\alpha^9)^2 + \alpha^{11}\alpha^5 + \alpha^{11} = \alpha^3 + \alpha^5 + \alpha^4 = x^3 + x + x^2 + x + x^2 + x^3 = 0$$

Лекция 9